Plotting points on a grid

In this sheet you will...

- describe a position with two numbers
- read and write coordinates (x, y)
- plot points in all four quadrants
- find a missing corner of a shape

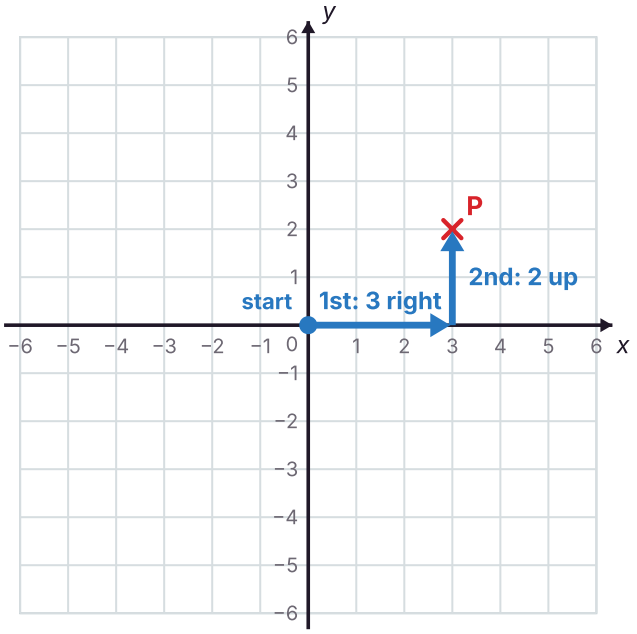
Key words

axis
origin
coordinates
quadrant

This sheet follows our lessons step by step, so you can read it again at home.

1 The coordinate plane

A grid is made of **two number lines crossing**: one horizontal and one vertical. They are called **axes**. They cross at the **origin**, which is the point 0 on both lines. A point on the grid needs **two numbers**: one to go along, one to go up or down.



x-axis: the horizontal line. Right is positive, left is negative.

y-axis: the vertical line. Up is positive, down is negative.

Origin: where the axes cross. Always start here.

To reach P: 3 along to the right, then 2 up. We write P(3, 2).

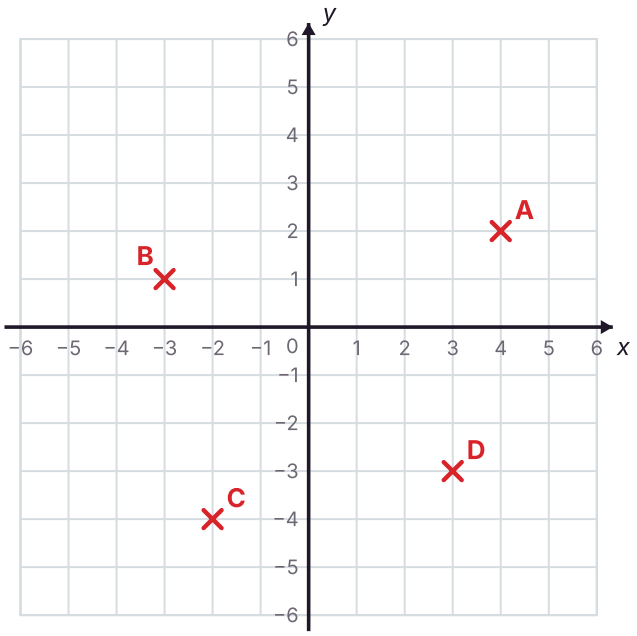
Key idea **Coordinates are written (x, y): first along, then up or down.**

Easy way to remember Along the corridor first, then up or down the stairs.

2 Reading and writing coordinates

Worked example 40.1

Write the coordinates of the points A, B, C and D.



A: 4 along, 2 up

Start at the origin. Right 4, up 2.

A(4, 2)

B: 3 left, 1 up

Left means a negative x.

B(−3, 1)

C: 2 left, 4 down

Both numbers are negative.

C(−2, −4)

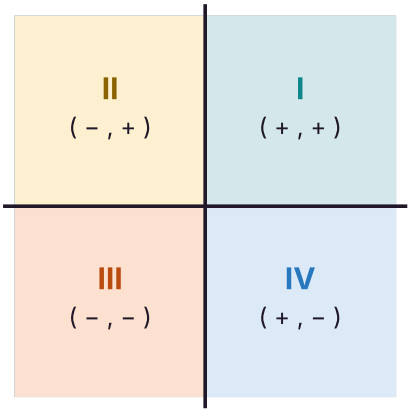
D: 3 right, 3 down

Down means a negative y.

D(3, −3)

3 The four quadrants

The axes cut the plane into four parts, called **quadrants**. We number them anticlockwise, starting at the top right.



The signs of x and y tell you the quadrant. In the second quadrant, x is negative and y is positive.

A point on an axis is in no quadrant: one of its coordinates is 0. Example: (0, −3) is on the y-axis.

Watch out!

The order matters. (3, −1) and (−1, 3) are different points. First along (x), then up or down (y).

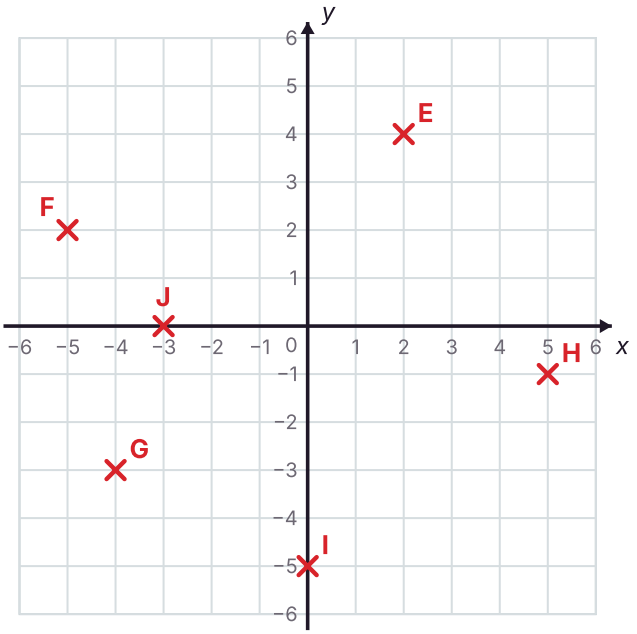
Do not count the origin. Start at 0 and count the steps.

Exercise 40.1

1 Write the coordinates of the points E to J.

Point	E	F	G	H	I	J
(x, y)						
Quadrant						

Write "I", "II", "III", "IV", or "axis" for the quadrant.

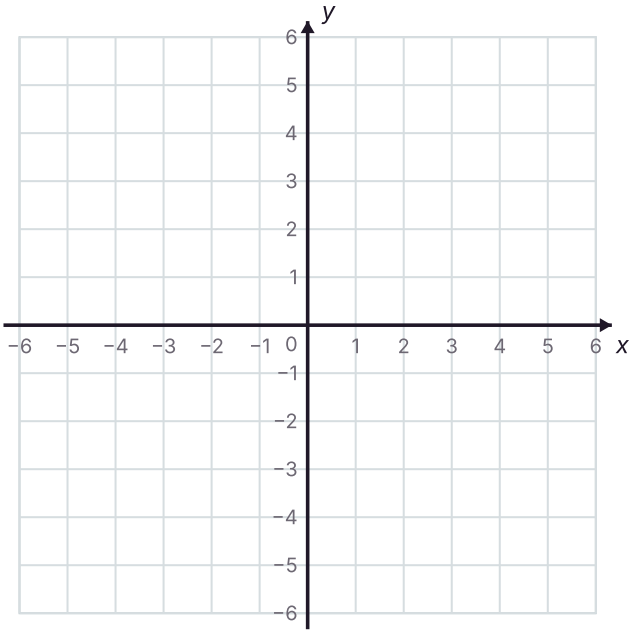


2 Plot these points on the grid and label them. Write the quadrant of each.

P(2, 5) Q(−3, 4) R(−5, −2) S(4, −4) T(0, −3)

P: ..... Q: ..... R: .....

S: ..... T: .....



3 Ines plots the point (−4, 2) by moving 2 to the left and 4 up. What mistake did she make? Where should the point be?

.....

.....

4 K(−4, −1), L(2, −1) and M(2, 3) are three corners of a rectangle KLMN.

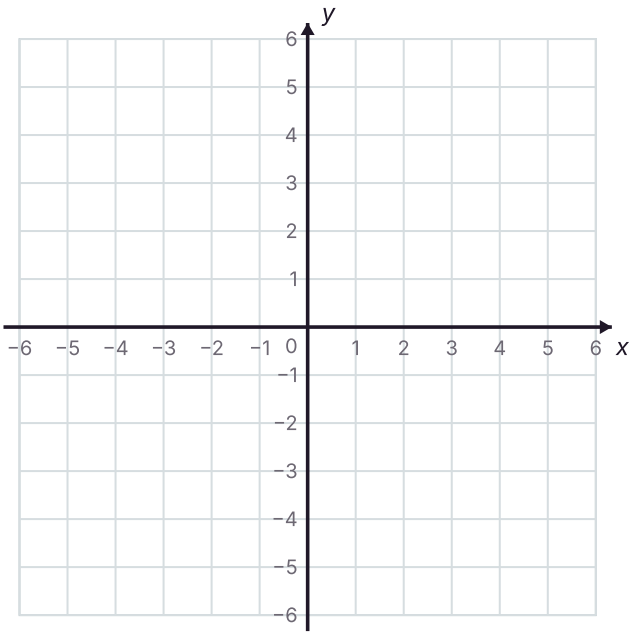
a) Plot K, L and M on the grid.

b) Find the fourth corner N without drawing it. Explain how you know.

.....

.....

c) Plot N and draw the rectangle. N = ...



What do you think?

Why is (3, 2) not the same point as (2, 3)?

Where can a point be if its *x*-coordinate is 0?

Exercise 40.2

1 Distance from the axes. Complete the table. The first row is done.

Point	Distance from the y-axis	Distance from the x-axis
(−5, 3)	5	3
(6, −2)		
(−4, −4)		
(0, 5)		
(−3, 0)		

Tip Distances are never negative. The distance of (−5, 3) from the y-axis is $|-5| = 5$.

2 Mystery points. Write the coordinates.

a) In quadrant II, 3 from the y-axis and 6 from the x-axis. .....

b) In quadrant III, 4 from the y-axis and 2 from the x-axis. .....

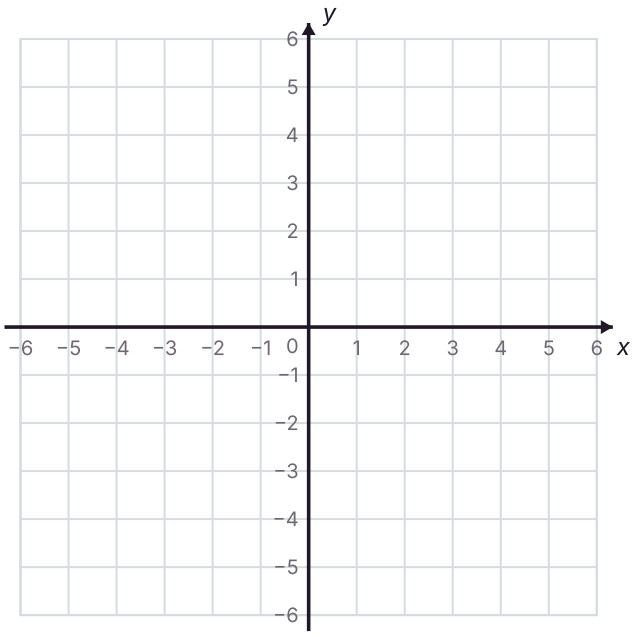
c) On the x-axis and 5 from the y-axis. There are two possible points. ....

3 Three corners of a parallelogram are (−4, −1), (0, −1) and (2, 3).

a) Plot the three points.

b) Find the fourth corner and plot it. .....

c) Draw the parallelogram.

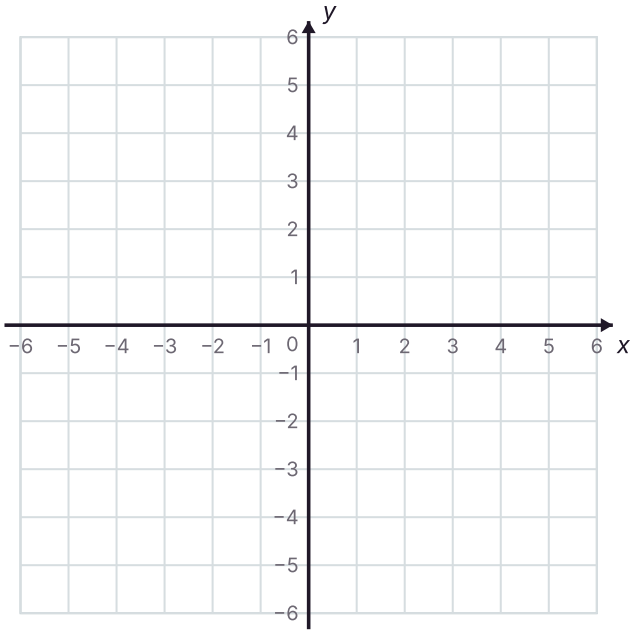


CHALLENGE

Plot these points and join them in order with a ruler:

(−4, −3), (−4, 1), (0, 4), (4, 1), (4, −3), back to (−4, −3).

Then plot and join: (−1, −3), (−1, −1), (1, −1), (1, −3). What do you see?



Summary checklist

- ☐ I know that coordinates are written (*x*, *y*): first along, then up or down.
- ☐ I can read the coordinates of a point and plot a point from its coordinates.
- ☐ I know the four quadrants and which signs each one has.
- ☐ I can find the distance of a point from each axis.
- ☐ I can find a missing corner of a rectangle or parallelogram.